1. Introduction

Educational systems are typically designed using standardized teaching methods, which often do not accommodate the diverse cognitive styles of neurodivergent students, including individuals with Autism Spectrum Disorder (ASD), Attention Deficit Hyperactivity Disorder (ADHD), dyslexia, and related learning differences. These students may experience challenges in attention, language processing, task organization, and communication, which can negatively affect academic performance and confidence.

The Cognitive AI Assistant for Neurodivergent Students is an AI-powered learning support system designed to act as a personalized academic assistant. The system uses Natural Language Processing (NLP), speech technologies, and machine learning to explain lessons in simple language, break tasks into smaller steps, adapt to individual learning styles, and provide multimodal support such as text, voice, and visuals. The assistant also maintains a memory of user preferences and learning history to deliver continuous personalization.

This Software Design Document (SDD) describes the overall system architecture, component-level design, interfaces, and data flow. The design focuses on accessibility, personalization, scalability through cloud infrastructure, and support for diverse communication needs.

2. Overall Description

2.1 Product Perspective

The system is a cloud-based AI learning assistant that operates as a distributed application:

Frontend: Web-based interface (React.js)

Backend: Python (Flask/Django) or Node.js with REST APIs

AI Services: NLP models, speech recognition, and text-to-speech

Cloud Infrastructure: Google Cloud Platform (GCP) for hosting and storage.

It functions as a support layer alongside existing educational content rather than replacing teachers.

2.2 Product Functions

The system provides the following key functions:

Lesson Simplification – Converts complex academic content into clear, easy-to-understand explanations.

Task Breakdown – Divides large assignments into smaller, manageable steps.

Adaptive Learning Support – Adjusts explanations based on student learning preferences and history.

Multimodal Assistance – Offers text, speech (STT/TTS), and visual learning aids.

Student Profile Memory – Stores preferences, strengths, and past interactions for personalization.

2.3 User Classes

Students (Primary Users) – Neurodivergent learners using the assistant for academic help.

Teachers – May monitor progress or customize learning material.

Parents/Guardians – View learning support usage and improvements.

2.4 Operating Environment

Platform: Web browser (desktop, tablet, mobile)

Cloud: Google Cloud Platform (GCP)

Internet: Required for AI model access and data sync

Accessibility: Designed following WCAG 2.1 accessibility guidelines

2.5 Design Constraints

Must support accessibility needs (keyboard navigation, readable fonts, contrast).

Must ensure secure storage of student learning data.

AI explanations must remain age-appropriate and educationally safe.

3. System Design

Overview

The system uses AI to process learning queries, generate simplified explanations, and provide personalized assistance based on stored student profiles.

Architecture

The system follows a client–server architecture:

Frontend (React.js)

User interaction interface

Displays explanations, steps, and visual aids

Backend Server (Python/Node.js)

Handles API requests

Manages user data

Connects to AI services

AI Module

NLP models (GPT, BERT, T5 via Hugging Face)

Speech-to-Text (Google Cloud STT)

Text-to-Speech (Google Cloud TTS)

Database & Storage

Stores student profiles, preferences, and interaction history

Hosted on GCP cloud storage/databases.

4. Component-Level Design

4.1 Frontend Component (React.js)

Subcomponents

User Interface Module – Chat-style interaction, lesson display, step-by-step guidance.

Accessibility Module – Font scaling, contrast mode, keyboard navigation.

Input Handler – Accepts text, voice, or selection-based inputs.

API Communication Module – Sends requests and receives AI responses.

4.2 Backend Server Component

API Controller – Manages requests from frontend.

Student Profile Manager – Stores and retrieves preferences and history.

Learning Adaptation Logic – Adjusts difficulty and explanation style.

Security Module – Authentication, authorization, data protection.

4.3 AI Processing Component

NLP Engine – Simplifies content and answers student queries.

Task Breakdown Engine – Converts tasks into smaller steps.

Speech Processing Module – Converts speech to text and text to speech.

Personalization Model – Learns patterns from past interactions.

4.4 Database Component

User Profile Storage

Interaction History Storage

Learning Preferences Storage

5. Interface Design

5.1 User Interface

Chat-style assistant interface

Icons and visual cues for easy understanding

Voice input/output options

5.2 Software Interfaces

NLP APIs – Hugging Face models

Speech APIs – Google Cloud STT & TTS

Cloud Services – GCP hosting and storage

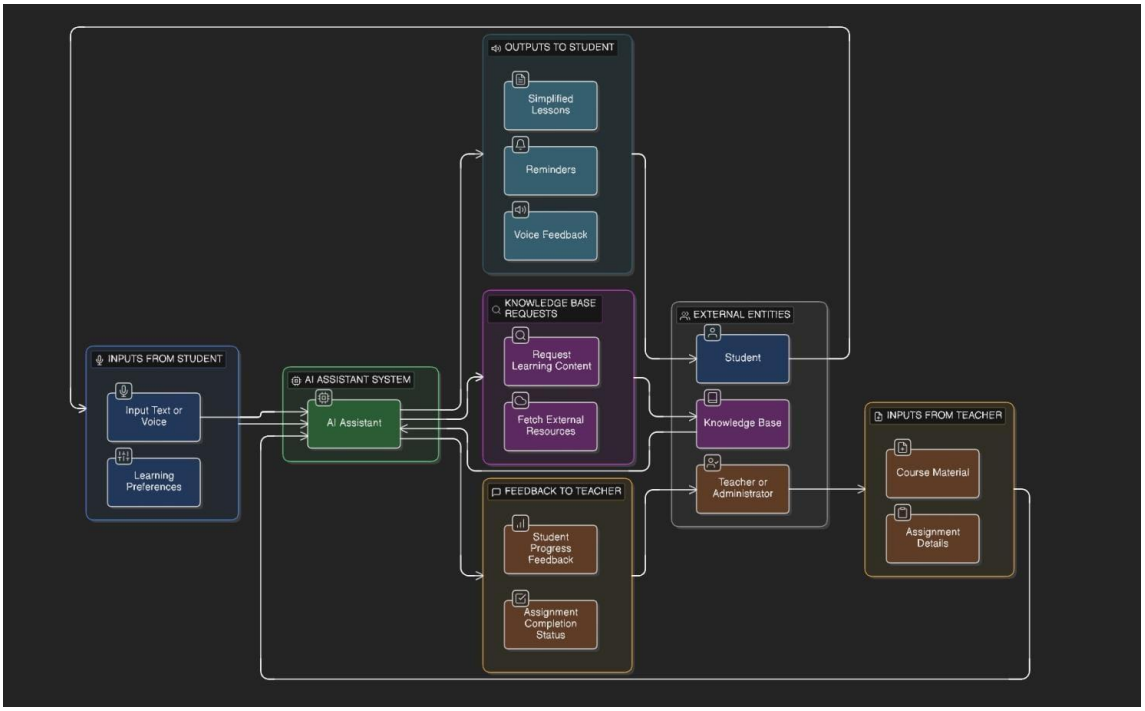
5.3 Communication Interfaces

HTTPS protocol

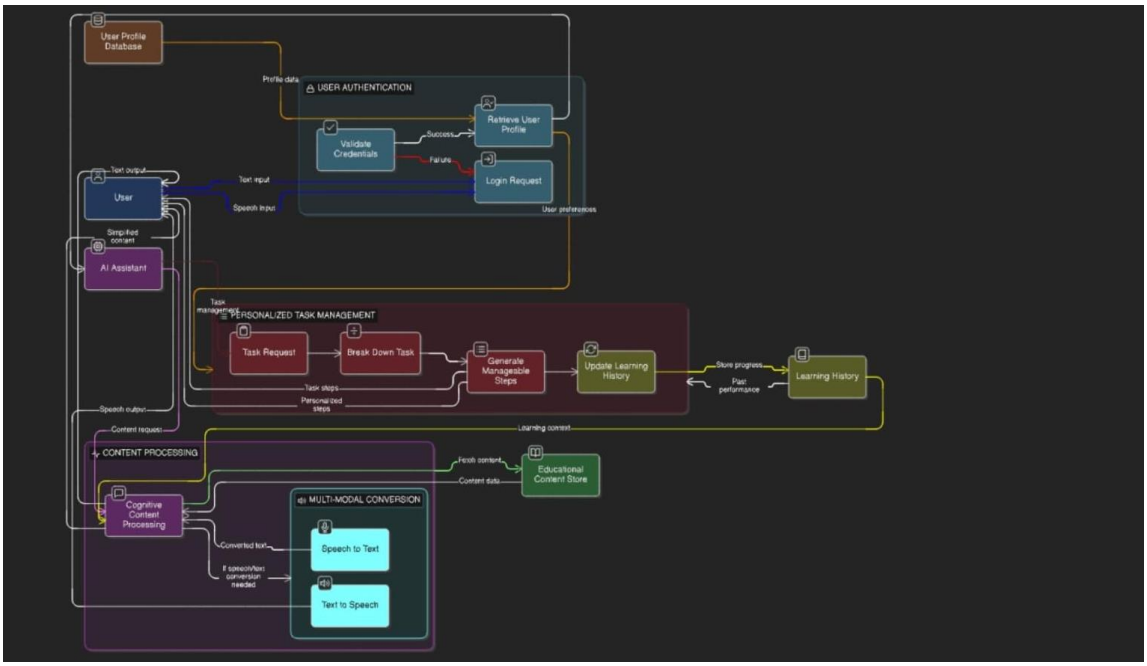
JSON data exchange format

6. Data Flow Description

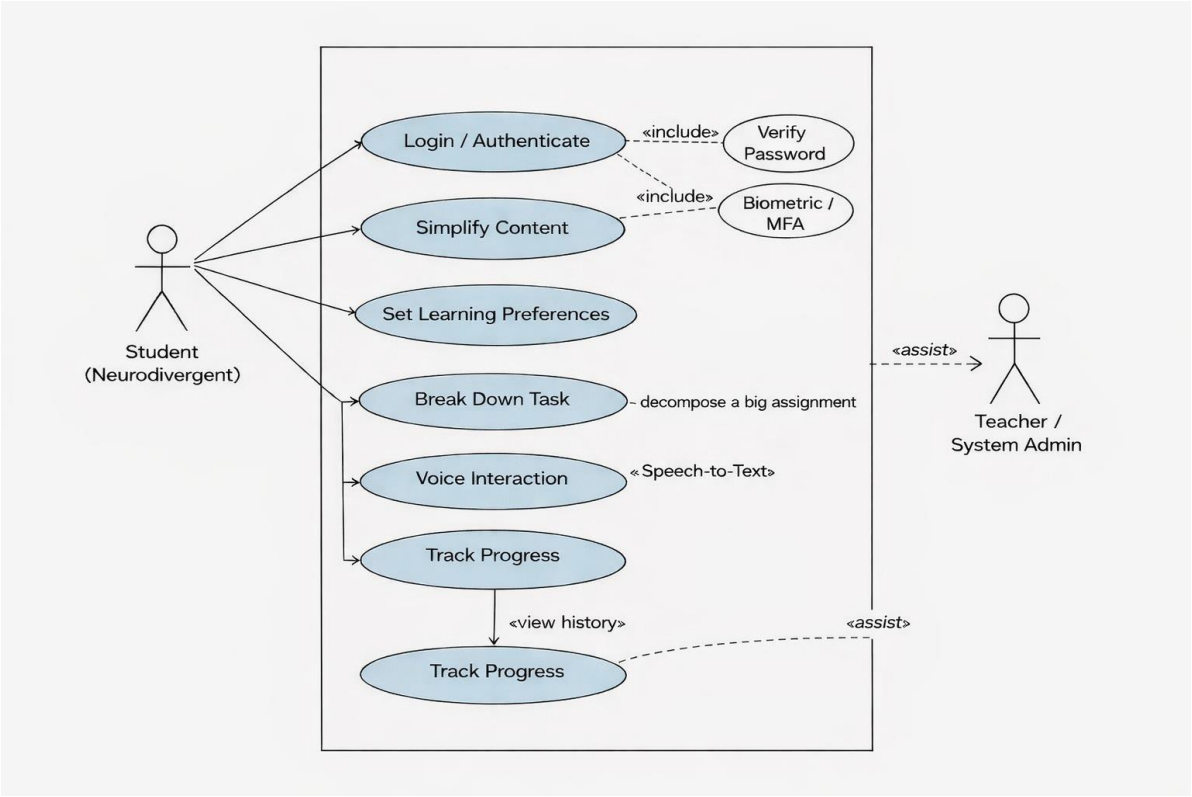
Level 0



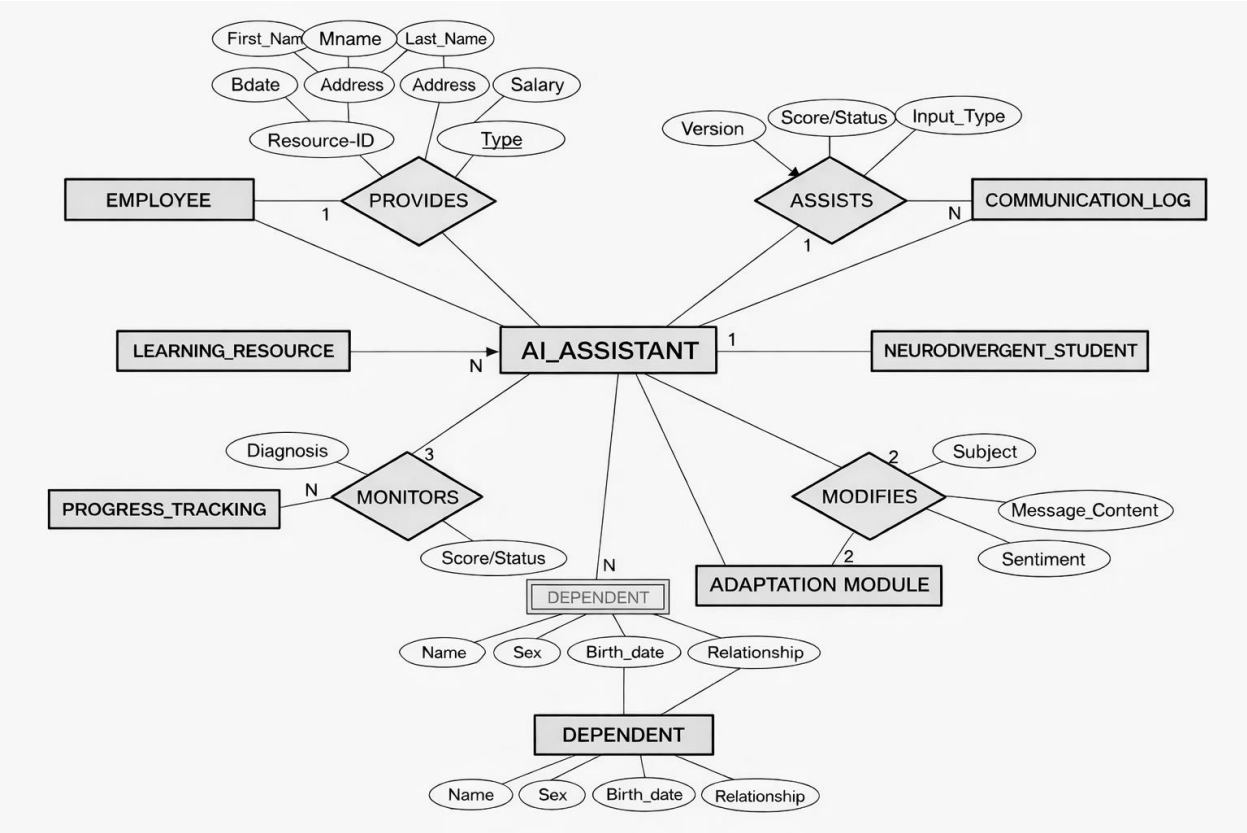
Level 1



7. UML Diagram



8. ER Diagram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